

Seeing space and time together: glyph maps in R

Line, interval and interactive glyph maps in ggplot2 with **cubble** and **sugarglider**

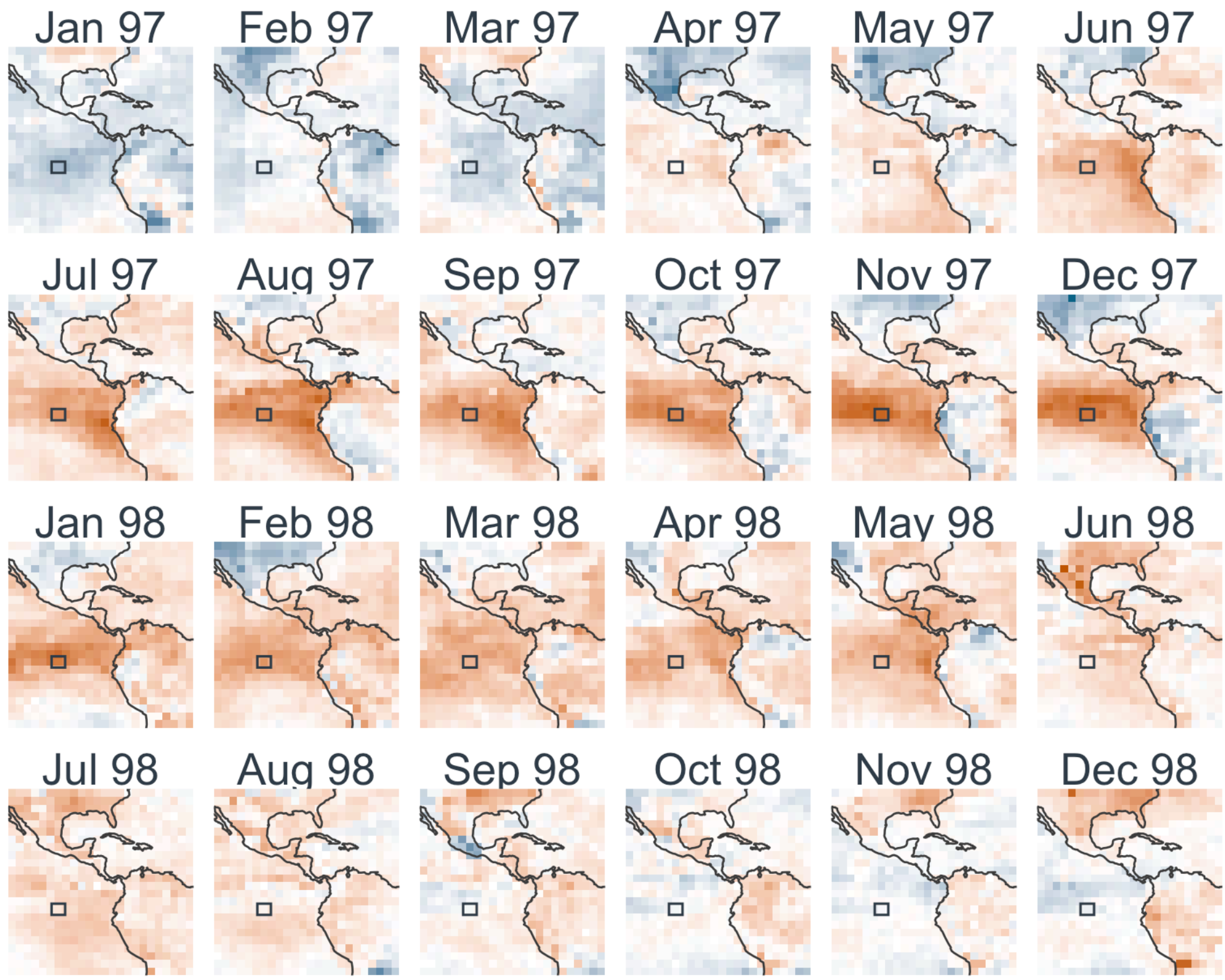
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2012 From facets to a glyph map

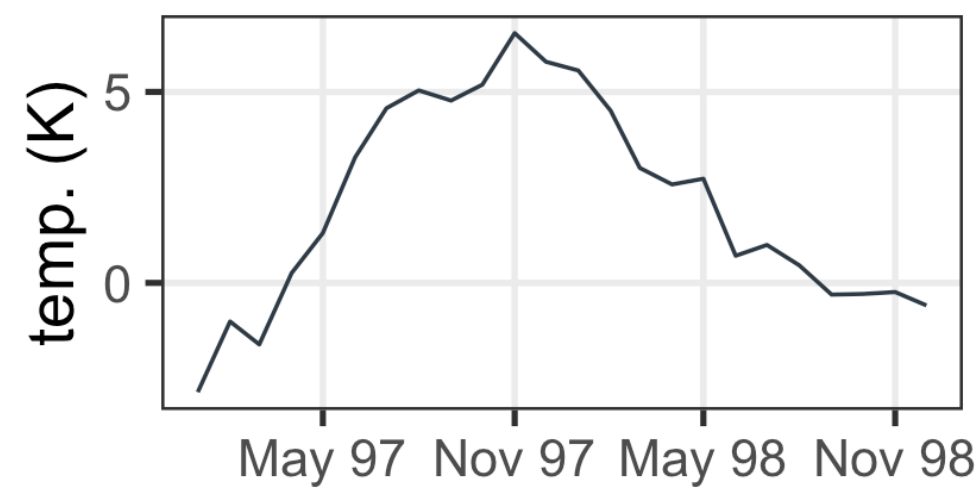
Environmental data often come as **many time series at many locations**, commonly shown as maps faceted by time. Below, monthly de-seasonalised surface temperature over Central America for 1997–98.



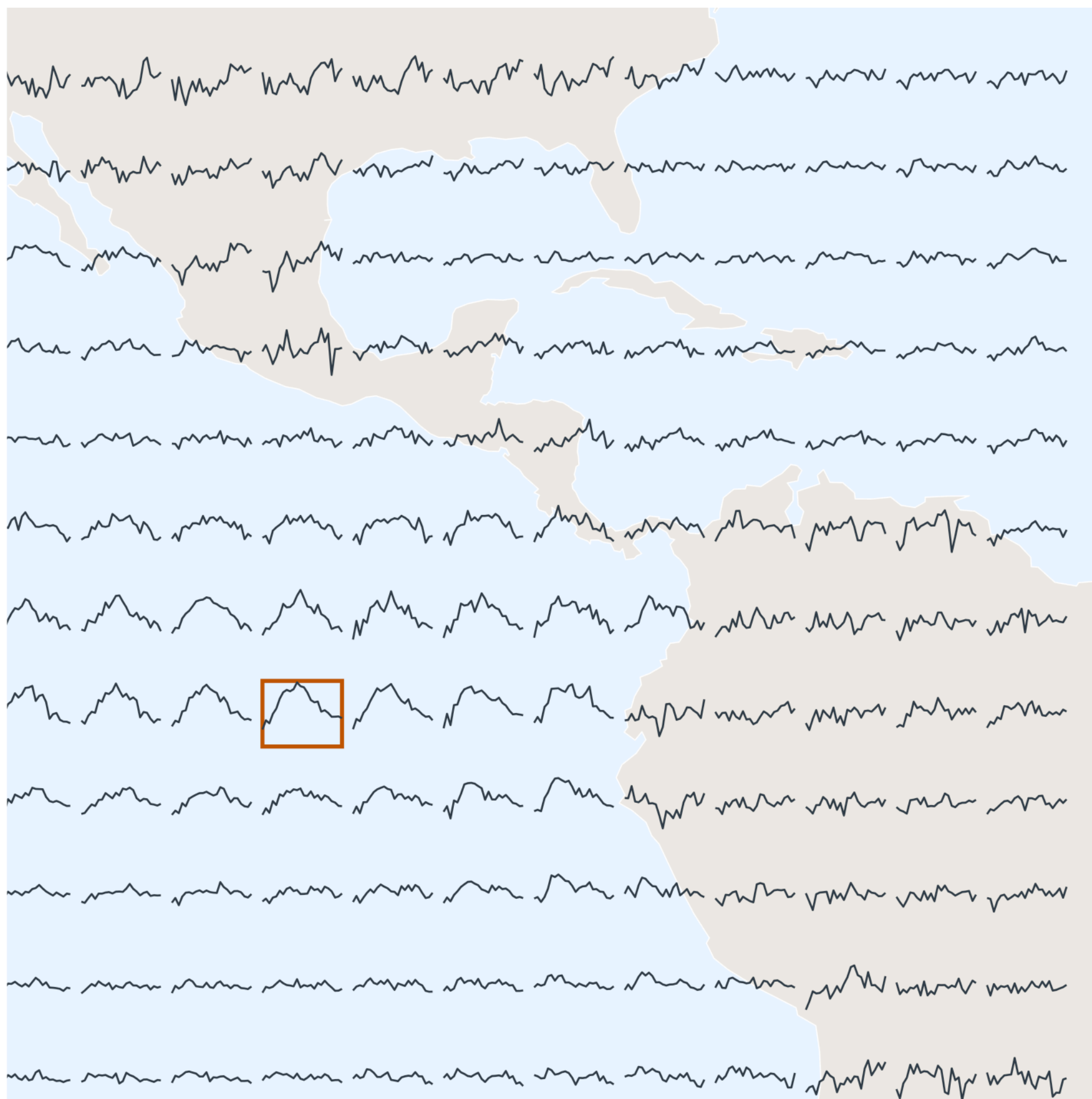
De-seasonalised temperature (K)

-4 0 4

Trace one location. Follow the boxed cell through the 24 panels to get its time series. A **glyph map** draws that series as a small plot at the cell's location (Wickham et al., 2012).



Do this for every location to get the full glyph map (`GGally::glyphs()` computes the glyph coordinates):



What the glyph map shows:

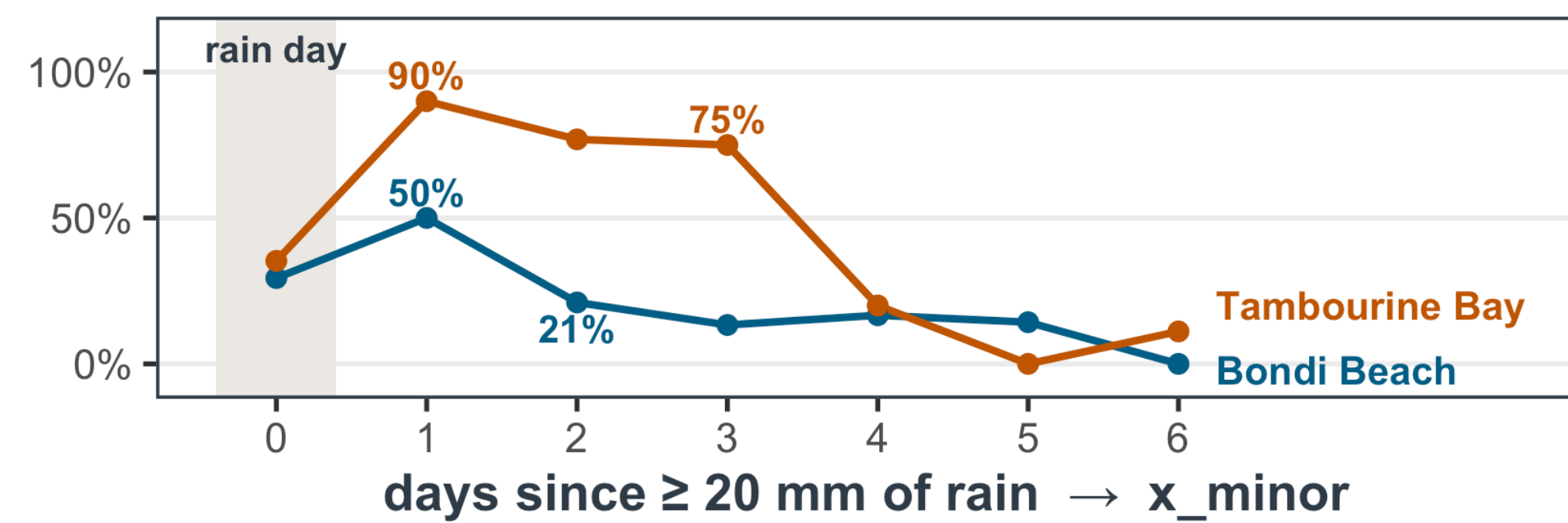
- Equatorial eastern Pacific glyphs share one hump peaking in late 1997 (the El Niño), fading away from the equator.
- Glyphs over land vary far more month to month than those over the sea.

2024 cubble: glyph maps in ggplot2

how long after a storm is it safe to swim in Sydney?

Each water sample (2015–2025) is matched to the most recent day with ≥ 20 mm of rain (a “very heavy rain” day in the ETCCDI climate-extremes indices), giving the number of days since rain when it was taken (0–6). For each site and each day, we take the percentage of samples above the bacteria guideline: 40 enterococci/100 mL, the limit for the best water-quality grade in Australia’s recreational-water guidelines (NHMRC, 2008) (TidyTuesday 2025-05-20).

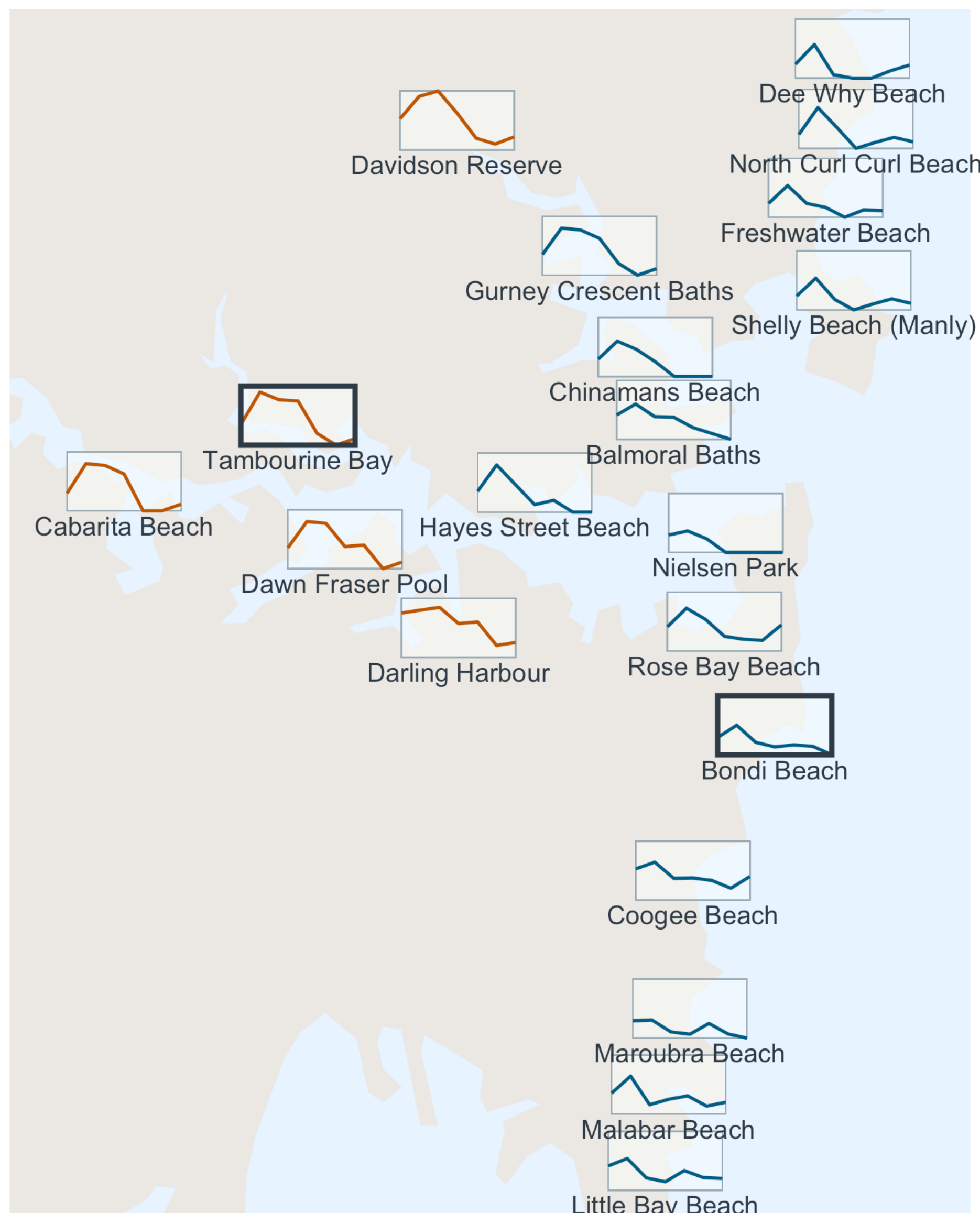
% of samples above guideline → y_minor



The two boxed glyphs below, full size: **Tambourine Bay** (inner harbour) stays polluted for three days; **Bondi Beach** clears by day 2.

`geom_glyph()` is a ggplot2 layer, so it combines with other layers (a base map, site names), scales and themes:

```
ggplot(beach, aes(x_major = long, y_major = lat,
                  x_minor = days, y_minor = percentage)) +
  geom_sf(..., inherit.aes = FALSE) + # base map
  geom_glyph_box(...) + geom_glyph(...) + # cubble glyphs
  geom_text(...) + # site names
  theme(...) # theme
```



Inner harbour Ocean & outer harbour

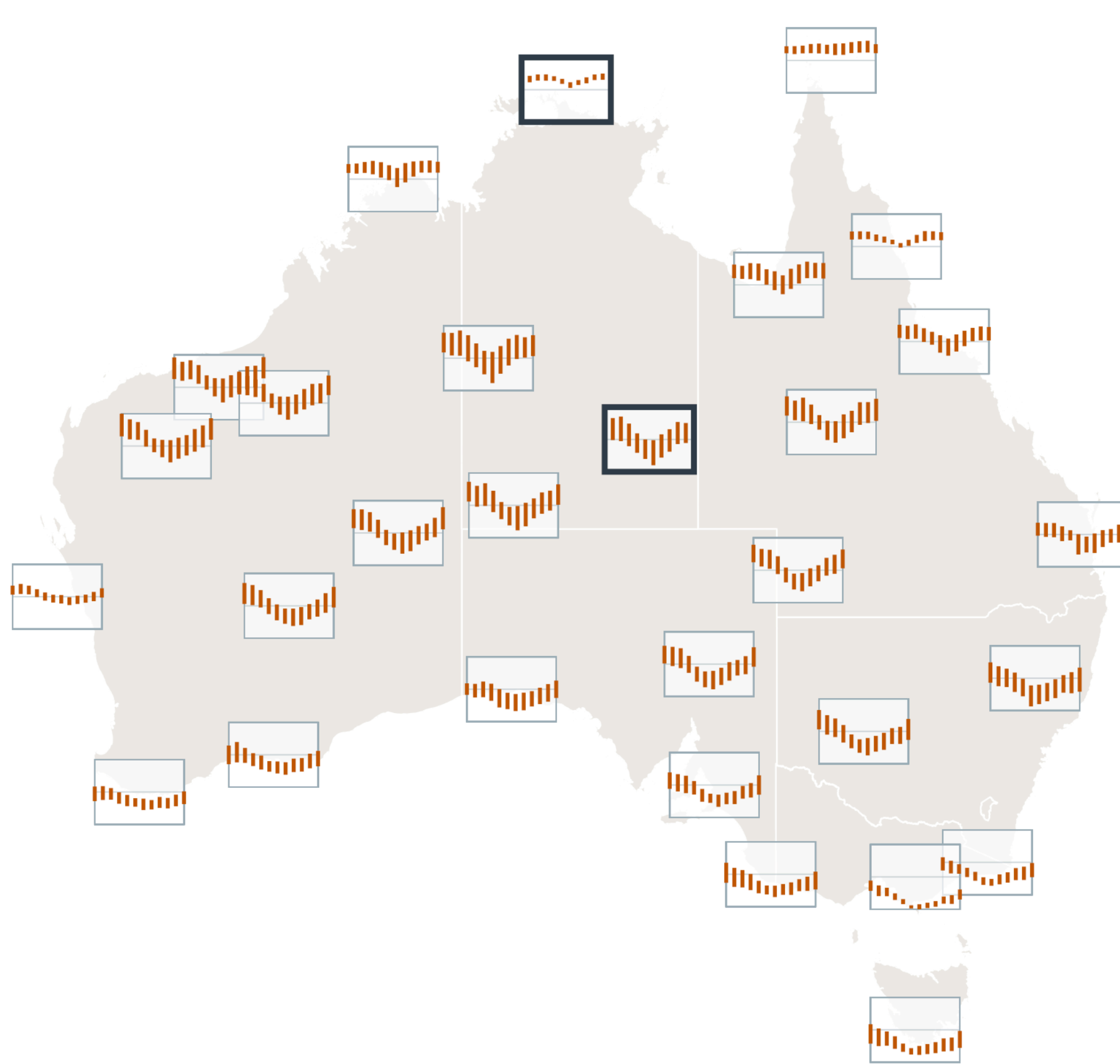
- Ocean & outer harbour** clear up by day 2 after a storm while **Inner harbour** needs until day 4–5.
- Coogee** and **Rose Bay**, graded Poor by NSW Beachwatch, stay high.

2026 sugarglider: interval segments

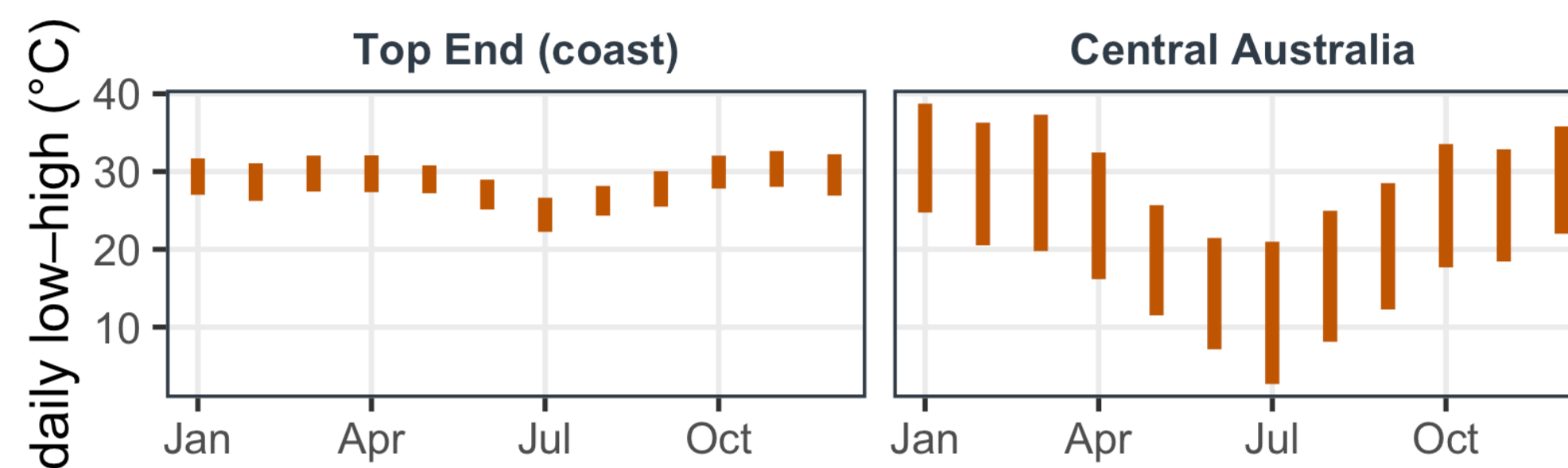
A line glyph shows **one value** per time point, but many summaries are an **interval**: a low and a high value of the same measurement, such as the daily minimum and maximum temperature.

`geom_glyph_segment()` draws one bar per time point. Here: the monthly average daily low to high temperature at Australian weather stations, 2020.

```
aus_temp |>
  ggplot(aes(x_major = long, y_major = lat, x_minor = month,
            y_minor = tmin, yend_minor = tmax)) +
  add_glyph_boxes() + geom_glyph_segment() +
  ...
```



The two boxed glyphs enlarged:



The **Top End** stays between 22 and 33 °C all year, with only 4–5 °C between day and night. **Central Australia** swings from 3 °C winter nights to 39 °C summer days, with 14–18 °C between day and night.

Take-home

- Glyph maps show space and time together.** Each location’s time series is drawn where it belongs, so local patterns and their spatial structure can be explored in a single plot during exploratory analysis.
- Glyph maps are now part of ggplot2.** cubble provides line glyphs and sugarglider adds interval glyphs (segments and ribbons), and both combine with base maps, scales, colour and facets like any other layer.
- Glyph maps can be interactive.** The same glyphs can be placed on a leaflet map, so viewers can zoom into dense areas and look up individual locations.

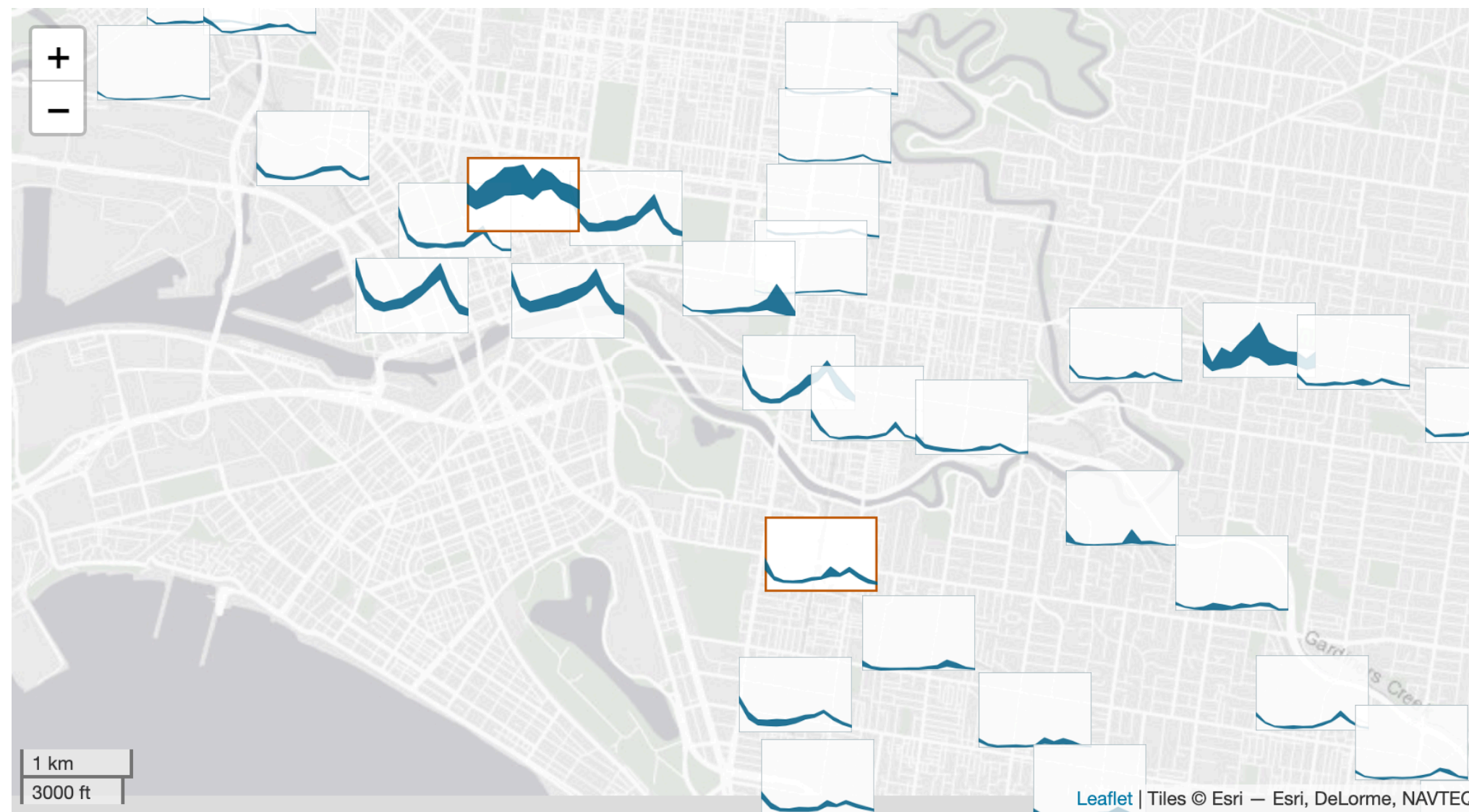
2026 sugarglider: interval ribbons

Ribbons fill the band between the lower and upper values. Melbourne train stations, hourly weekday passengers, 2023–24 (`sugarglider::train`), on an interactive **leaflet** map. For each hour, the **lower edge** is the passenger count in the quietest month and the **upper edge** in the busiest month. Each station’s glyph is saved as an image and added as a leaflet marker (simplified):

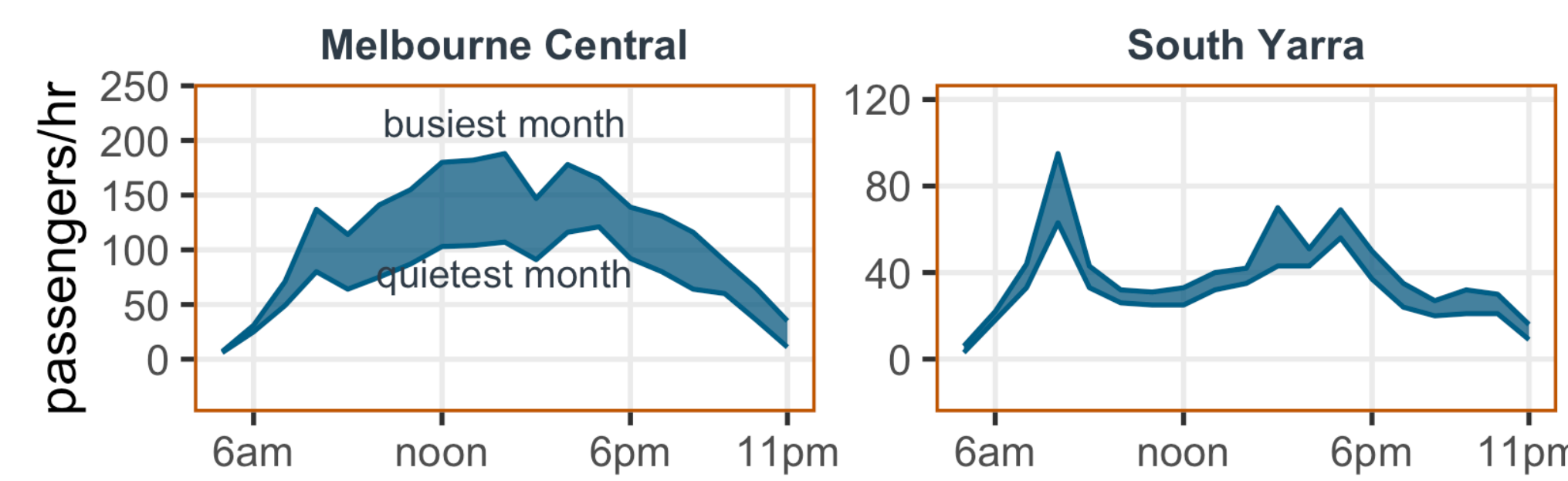
```
glyph <- ggplot(station,
  aes(x_major = long, y_major = lat, x_minor = hour,
      ymin_minor = min_weekday, ymax_minor = max_weekday)) +
  add_glyph_boxes() + geom_glyph_ribbon() + theme_void()
ggsave("glyph.png", glyph, bg = "transparent")

leaflet() |> addProviderTiles("Esri.WorldGrayCanvas") |>
  addMarkers(lng, lat, label = name,
    icon = makeIcon("glyph.png", 66, 44))
```

All stations share one scale, so glyph height shows how busy a station is; zoom in to separate crowded glyphs and hover for station names:



The two boxed glyphs enlarged, with their own axes:



City Loop stations carry by far the most passengers. On their own axes, suburban stations such as **South Yarra** show two sharp **commuter peaks**, while **Melbourne Central** is as busy at midday as at the peaks.

Questions? Email hsherryzhang@utexas.edu



Wickham, Hofmann, Wickham & Cook (2012). Glyph-maps for visually exploring temporal patterns in climate data and models. *Environmetrics* 23(5), 382–393. [doi:10.1002/env.2152](https://doi.org/10.1002/env.2152)
Zhang, Cook, Laa, Langrené & Menéndez (2024). cubble: An R package for organizing and wrangling multivariate spatio-temporal data. *J. Stat. Softw.* 110(7). [doi:10.18637/jss.v110.i07](https://doi.org/10.18637/jss.v110.i07)
Po, Yang, Zhang & Cook (2026). sugarglider: Create glyph-maps of spatiotemporal data. *The R Journal* 18(3), 286–304. [doi:10.32614/RJ-2026-050](https://doi.org/10.32614/RJ-2026-050)
Leaflet tiles: Esri. Beach data: [TidyTuesday 2025-05-20](https://tidytuesday.com/2025/05/20).